

Biyaherong Chess Coach goes fully offline

Rebuilding the app for iPhone and iPad so that every lesson, puzzle and game works without an internet connection — and without a monthly server bill.

Works on the plane · Works in the province · Works forever

The idea in one page

Today the app is really three products stitched together: the phone app, a rented server that stores everything, and a second rented computer that thinks about chess moves. If any of those is switched off — or if the player has no signal — the app stops working.

We are moving all of it **onto the phone itself**. The chess engine, the puzzle library, the lessons and the player's progress will all live inside the app. The result is an app that opens instantly, works anywhere in the Philippines regardless of signal, and costs nothing to run once it is built.

550,000	7	₱0	1x
PUZZLES INSIDE THE APP	FEATURES CARRIED OVER	MONTHLY SERVER COST	PAYMENT, NOT MONTHLY

The single rule behind every decision in this document

The finished app must run with **no service that charges us monthly**. A feature either works completely on the phone, or it is a small bonus that appears only when the player happens to be online. Nothing in the app is allowed to break when the signal drops.

What the player will notice

TODAY	AFTER THE REBUILD
Must create an account, wait for an email code, and log in	Opens straight to the home screen — no account, no password, no email
Puzzles and analysis freeze when the signal is weak	Everything responds instantly, even in airplane mode
Engine analysis takes a moment to travel to our server and back	Analysis appears immediately — the engine is inside the phone
Can only be logged in on one device at a time	No such restriction — the app simply belongs to the phone
Pays every month, and loses access when payment stops	Pays once, and keeps the app for life

What the app will do

All seven of the features that players actually use every day are carried over in full. Nothing about how they work on screen changes — they simply keep working when there is no signal.

1 Puzzle Hub

All five puzzle modes stay: **Play Puzzles** (rated, your rating goes up and down), **Daily Puzzle**, **Thematic Puzzles** (forks, pins, back-rank mates and nine more), **Puzzle Turbo** and **Puzzle Streak**. Over half a million puzzles will be stored inside the app.

2 Analysis Board

The full study board — set up any position, import a game, explore variations, and see what the engine thinks of every move. The evaluation bar and graph stay. This becomes **noticeably faster**, because there is no longer a trip to our server for every move.

3 Game Review

After any game, the app grades every move — Brilliant, Best, Good, Inaccuracy, Mistake, Blunder — and gives each player an accuracy score, with the evaluation graph of the whole game. Currently limited to a few reviews per hour; offline it becomes **unlimited**.

4 Play vs Coach

All five coach characters keep their personalities and playing strength — Jaden Pogi, Pretty Jade, Handsome Jude, Mommy Julie and Coach Pogi — each with their own favourite openings. Take-back and premove stay. Unfinished games are saved automatically.

5 Openings Trainer

The Chessable-style memorisation system with 22 ready-made opening repertoires. The app remembers which positions you find hard and brings them back at the right moment. Players can still upload their own openings as a PGN file.

6 Tournament Manager

Swiss and Round Robin pairings, results entry, standings with proper tiebreaks, and shareable pairing sheets. This is the feature **local organisers** use at real tournaments — usually in venues with poor signal, which is exactly where offline matters most.

7 Profile & Statistics

Puzzle rating, best rating, accuracy, solving activity for the week, improvement over time, and a breakdown of which tactical themes you are strong or weak at. Board colour themes stay too.

Bonus Master Games

The grandmaster games database from Lichess stays in the app. It is the one feature that needs internet — but it is **completely free to use** and costs us nothing. When there is no signal the panel simply stays quiet; nothing else is affected.

Two things get better, not just equal

- **Speed.** Every engine answer today travels to a server and back. On the phone it is instant, so the analysis board and coach games feel considerably more responsive.
- **No more limits.** The caps that exist today — 30 analyses per minute, 5 game reviews per hour — exist only to protect our server from cost. They disappear entirely.

What we are letting go

Some features only work because a server is holding everyone's information in one place. Those cannot survive the move, and it is better to say so plainly than to ship a half-working version.

FEATURE	WHY IT CANNOT COME ALONG	WHAT REPLACES IT
Leaderboards	Ranking players against each other needs one shared list that everyone reads from	Personal best scores and a share card so players can post their results themselves
Social features search, follow, public profiles	Same reason — there is no shared place for players to find each other	Nothing. <i>This is the one real loss.</i>
Tutorial videos	Video hosting is charged per gigabyte watched, and putting hours of video in the app would make the download enormous	A link that opens the videos on YouTube or Facebook — free to host, and the content stays available
Accounts & login registration, email codes, password reset	All of it requires a server and a paid email service	The app opens immediately. Progress is backed up through the player's own iCloud.
Opening Tree the older "My Tree" feature	It downloads hundreds of the player's own online games each time	The Openings Trainer, which does the same job better
Shareable analysis links	A public web link needs a website to point at	Sharing the game file or a picture of the board through the normal share button
Web admin panel	There is no longer a server for it to manage	Not needed

The honest summary

What we genuinely lose is **social comparison** — leaderboards and following other players — and **in-app video lessons**. Everything else is either kept as-is or improved. If competing against other players becomes important later, it can be added back as a separate online extra without disturbing the offline app.

The business case

WHAT WE PAY FOR TODAY	TODAY	AFTER THE REBUILD
Application server	Rented monthly	Removed
Database (all player data)	Rented monthly	Removed
Chess engine computer	Rented monthly	Runs on the phone
File storage for videos and photos	Charged per use	Removed
Email service for verification codes	Charged per message	Removed
Domain, security certificate, monitoring	Yearly + upkeep	Removed

WHAT WE PAY FOR TODAY	TODAY	AFTER THE REBUILD
Apple Developer Program	—	\$99 / year
Ongoing cost to keep the app alive	Grows with every new user	\$99 / year — flat, forever

Why the monthly cost matters more than it looks

Under the current setup, **every new user makes the bill bigger** — more puzzles served, more engine time, more storage. Success is expensive. After the rebuild, ten users and ten thousand users cost exactly the same to support: nothing. Growth becomes pure upside.

Moving from monthly subscription to one-time payment

Since there is no server to keep paying for, there is no reason to keep charging players every month. The app becomes a **single purchase** through the App Store — the player pays once and owns it permanently, on every Apple device they own.

	MONTHLY SUBSCRIPTION	ONE-TIME PURCHASE
For the player	Cheaper to start, but never stops; access is lost if they stop paying	Costs more once, then never again — easier to say yes to
For us	Predictable income, but constant pressure to justify the next payment	Income at the moment of sale; no cancellations, no billing support, no refund chasing
Fit for the market	Filipino players are noticeably resistant to recurring app charges	A far better fit — a one-time price reads as fair and final

The free version stays genuinely useful — five rated puzzles a day, the daily puzzle, two of the five coaches, and the analysis board. That is deliberate: with a one-time purchase, players buy after they have already felt the app is worth it.

How we get there

1 Decisions first

Two questions below need your answer before any building starts. Both are cheap to settle now and expensive to change later.

2 Foundations — the board and the storage

The chessboard, the rules of chess, and the place where all progress is stored. Nothing is visible to a player yet, but everything else is built on top of it.

3 The chess engine on the phone

The hardest and riskiest part, so we do it early. Once the engine runs well on a real iPhone without draining the battery, the rest of the project is predictable.

4 Puzzles, then Play vs Coach

The two features players use most. At the end of this stage the app is genuinely usable and can be put in front of real testers.

5 Analysis Board, Game Review, Openings, Tournaments

The deeper study tools. Tournaments can be built alongside the others, since it shares almost nothing with the rest of the app.

6 Profile, purchase, polish and App Store submission

Statistics, board themes, the purchase screen, the privacy policy, and a TestFlight round with real players before we submit for review.

Two decisions we need from you

1. How large should the app download be?

The complete puzzle library is very large. Fitting all 550,000 puzzles would make the app roughly **100 MB** to download — large enough that Apple warns users on mobile data, which costs us installs. Our recommendation is to include about **150,000 carefully chosen puzzles** instead, bringing the download to roughly **25 MB**. No single player will ever run out of puzzles at that number, and the app installs quickly even on a slow connection.

2. Which chess engine do we use?

The engine we use today, Stockfish, is free but carries a licence requiring that any app built around it also publishes its own source code. Several chess apps have been forced to comply after publishing on the App Store, so this is a real risk rather than a technicality. There are two safe paths: **publish our code openly**, or **use a different engine with a friendlier licence** — strong enough for coaching and analysis, with no such obligation. Our recommendation is the second, and we suggest confirming it with a lawyer before we start.

What happens to current subscribers

The offline app is a fresh start — ratings and history do not carry over, because there are no accounts to carry them from. We recommend giving every existing paying subscriber the new app **unlocked at no charge**. It costs us nothing, it protects goodwill, and it turns the switch-over into good news rather than a complaint.